13-10-2026

(outlook) OL	(Temperature) T	(Humidity) H	(windy) W	(Rain) R
sunny	Hot	High	weak	NO
sunny	mild	High	Strong	NO
cloudy	Hot	High	weak	Yes
Rainy	mild	High	weak	Yes
Rainy	cool	Normal	weak	Yes
Rainy	cool	Normal	Strong	NO
cloudy	Mild	Normal	strong	Yes
Sunny	cool	Normal	weak	NO

Antropy of prime attributes

Rain

Yes NO

$$\frac{4}{8}\log_2\frac{4}{8} - \frac{4}{8}\log_2\frac{4}{8}$$

$$-0.5\log_2(0.5) - 0.5\log_2(0.5)$$

$$-0.5\frac{\log(0.5)}{\log 2} - 0.5\frac{\log(0.5)}{\log 2}$$

$$-0.5 \times -1 - 0.5 \times -1$$

$$0.5 + 0.5$$

$$1$$

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If o again of outlook

	yes	NO	Entropy
Sunny	0	1	0.00
cloudy	2	0	0.00
Rainy	2	1	0.083

$$-\frac{2}{3}\log_2\frac{2}{3} - \frac{1}{3}\log_2\frac{1}{3}$$

$$-0.66\,\frac{\log(0.66)}{\log 2} - 0.33\,\frac{\log(0.33)}{\log 2}$$

$$-0.66 \times 0.59 - 0.33 \times -1.59$$

$$0.389 + 0.524$$

$$0.917$$

$$IG = 1 - 0.917$$

$$= 0.083 \qquad \text{weight} = 0.657$$

Temperature

	Yes	NO	Total	Entropy
Hot	1	1	2	1
mild	2	1	3	0.922
cool	1	2	3	0.922

Hot

$$-\frac{1}{2}\log_2\frac{1}{2} + -\frac{1}{2}\log_2\frac{1}{2}$$

$$-0.5\,\frac{\log(0.5)}{\log_2} + -0.5\,\frac{\log(0.5)}{\log_2}$$

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$-0.5 \times -1 \quad + \quad -0.5 \times -1$

$0.5 \quad + \quad 0.5$

1

mild

$$-\frac{2}{3}\log_2\frac{2}{3} + -\frac{1}{3}\log_2\frac{1}{3}$$

$$-0.66 \times \frac{\log(0.66)}{\log 2} + -0.33 \times \frac{\log(0.33)}{\log 2}$$

$-0.66 \times -0.594 + \quad -0.33 \times -1.594$

$0.395 + 0.527$

0.922

cool

$$-\frac{1}{3}\log_2\frac{1}{3} + -\frac{2}{3}\log_2\frac{2}{3}$$

$$-0.33\,\frac{\log(0.33)}{\log 2} + -0.66\,\frac{\log(0.66)}{\log 2}$$

$-0.33 \times -1.594 + \quad -0.66 \times -0.594$

$0.527 + 0.395$

0.922

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Humidity	Yes	NO	Total	Entropy
High	2	2	4	1
Normal	2	2	4	1

$$-\frac{2}{4}\log_2\frac{2}{4} + -\frac{2}{4}\log_2\frac{2}{4}$$

$$-0.5\,\frac{\log(0.5)}{\log 2} + -0.5\,\frac{\log(0.5)}{\log 2}$$

$$-0.5 \times -1 \quad + \quad -0.5 \times -1$$

$$0.5 \quad + 0.5$$

$$1$$

windy	Yes	NO	Total	Entropy
weak	3	2	5	0.948
strong	1	2	3	0.915

$$-\frac{3}{5}\log_2\left(\frac{3}{5}\right) + -\frac{2}{5}\log_2\left(\frac{2}{5}\right)$$

$$-0.6\,\frac{\log(0.6)}{\log 2} + -0.4\,\frac{\log(0.4)}{\log 2}$$

$$-0.6 \times -0.7 \quad + \quad -0.4 \times -1.32$$

$$0.42 \quad + 0.528$$

$$0.948$$

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$$-\frac{1}{3}\log_2\frac{1}{3} + -\frac{2}{3}\log_2\frac{2}{3}$$

$$-0.33\,\frac{\log(0.33)}{\log_2} + -0.66\,\frac{\log(0.66)}{\log_2}$$

$$-0.33 \times -1.599 + -0.66 \times -0.599$$

$$0.52 + 0.395$$

$$0.915$$

Rain 1

outlook 0.657

Temp 0.922

Humidity 0

The root node is 0.657

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